Commercial Data Science: Beyond Theory to Real Impact

For aspiring data scientists who know the theory but lack real-world experience

1

Chance to become part of the DS team for top performers

2

10 Real Banking Projects

3

2-3 Years of Equivalent DS Experience

Elective Course by Bota Abdraimova — Head of ML & AI, Anorbank

Data Science by the Numbers

Explore the promising landscape of Data Science careers, from explosive growth and attractive salaries to the evolving gender distribution.

Market & Growth

Projected job growth: **+34%** (2024–2034)

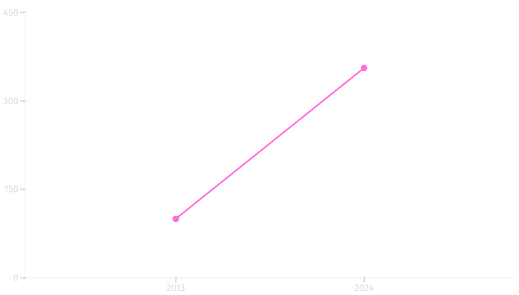
Average Salary

Median salary: **$120,000**/year
Range: **$78,000 – $193,000+**

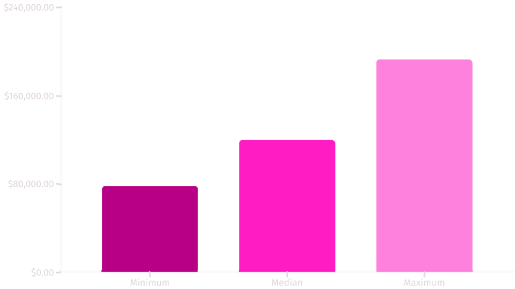
Gender Gap

Global: Women 15–26% of DS professionals

Job postings since 2013: **+256%**

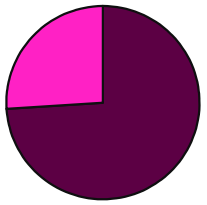


Sources: BLS.gov, Market.us, LinkedIn



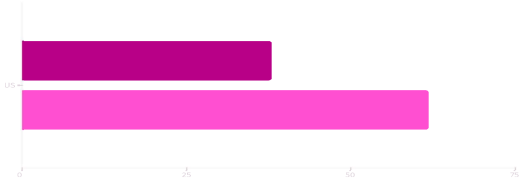
Sources: BLS.gov, Esparkinfo

US: ~38% women



Men (Global)

Women (Global)



Women (%)

Men (%)

Sources: Market.us, USD Online, CareerExplorer

What's Your Data Science Journey So Far?

Let's quickly check where everyone stands to tailor our discussion:

- Have you ever built a Kaggle project?
- Have you actually worked with real, messy business data?
- Who here already knows Python?
- Who here already knows SQL?
- Who here already knows Statistics?
- Who here wants to land a data science job right after graduation?

My Mission: Bridging the Gap

The goal is to equip you with the practical experience employers truly value.

The Theory-Practice Gap

Traditional education often leaves a disconnect between academic knowledge and real-world application.

Practical Experience You Need

Gain hands-on skills and build a portfolio of work that directly addresses industry challenges.

This Course: Your Bridge

We provide the essential link, transforming theoretical understanding into valuable, job-ready expertise.

Course Structure: 10 Real Banking Projects

1

Block 1: Foundations & Customer Insights

Banking analytics, SQL for data marts, feature engineering, and customer segmentation

2

Block 2: Classic ML for Marketing & Risk

Churn prediction, cross-sell modeling, next best offer systems, credit scoring, and collection strategy

3

Block 3: Advanced AI Applications

NLP for banking, conversational AI analytics, document intelligence with OCR, and graph-based fraud detection

4

Block 4: Portfolio & Presentation

Each week, you'll tackle a new case with real data and real problems, building a portfolio that showcases your ability to deliver business impact.

Data storytelling, dashboarding, and final portfolio presentations to demonstrate business acumen

The Reality Gap in Data Science Work

University Focus

Heavy emphasis on data preprocessing and algorithm theory

Industry Reality

50% - Problem definition & business translation
40% - Feature engineering & selection
10% - Data preprocessing

Your Career-Changing Outcomes

Beyond Classroom Projects

You'll build a portfolio equivalent to 2-3 years of real-world experience that demonstrates your ability to:

- Solve end-to-end business problems
- Apply advanced techniques to real banking data
- Communicate insights that drive decisions

Business-Oriented Thinking

Employers consistently value data scientists who can:

- Translate business questions into analytical frameworks
- Connect model outputs to revenue and cost impacts
- Present technical findings to non-technical stakeholders

You'll learn to think like a business-oriented Data Scientist — not just a theoretician, but someone who can make data science valuable for the company.

Who This Course Is For: Prerequisite Knowledge

This is not an introductory course. To truly thrive and gain the most from our practical projects, you need a solid foundation in the following core data science skills:

Python

Comfort with Python programming, including data structures, basic algorithms, and libraries like Pandas and NumPy for data manipulation.

SQL

Ability to write and understand SQL queries for data extraction, transformation, and analysis from relational databases.

Statistical Foundations

A grasp of fundamental statistical concepts, hypothesis testing, and common probability distributions.

Machine Learning Fundamentals

Familiarity with common machine learning models (e.g., linear regression, decision trees, clustering) and their basic principles.

While we won't be covering these basics, our focus will be on equipping you with the advanced mindset and professional tools to apply this knowledge to real-world banking challenges.

Ready to Bridge the Gap?

This elective is for those serious about becoming data scientists who can:

1 **Think like the business**

Apply data science to real banking challenges

2 **Build an outstanding portfolio**

Stand out with projects that mirror industry work

3 **Fast-track your career**

Skip years of on-the-job learning

Scan to secure your spot

Limited seats available!